COMP534 – Assessment 3

Deadline: 8th of May 2026

Chunking, also known as shallow parsing, is a subtask of Natural Language Processing (NLP) that involves identifying and segmenting text into syntactically meaningful units, such as noun phrases and verb phrases. Chunking plays a crucial role in many NLP applications (including information extraction, information retrieval, and question answering) by providing a shallow structural representation of sentences that supports more efficient and accurate language understanding. In this assignment, your aim is to build and train neural network models to solve this task.

Specifically, you will use a small dataset composed of around 8870 sentences, **which can be downloaded on Canvas.** Each sentence contains a variable number of words. In this dataset, each line has 2 columns (separated by a space delimiter): the first is the word and the second is the chunk tag of that word. There are 17 possible chunk tags:

1. "B-NP": Beginning Noun Phrase
2. "I-NP": Inside Noun Phrase
3. "B-ADVP": Beginning Adverb
4. "I-ADVP": Inside Adverb
5. "B-CONJP": Beginning Conjunction
6. "I-CONJP": Inside Conjunction
7. "B-ADJP": Beginning Adjective
8. "I-ADJP": Inside Adjective
9. "B-VP": Beginning Verb
10. "I-VP": Inside Verb
11. "B-PP": Beginning prepositional
12. "I-PP": Inside prepositional
13. "B-SBAR": Beginning subordinate clause
14. "I-SBAR": Inside subordinate clause
15. "B-INTJ": Beginning interjection
16. "B-PRT": Beginning particle
17. "O": outside

Sentences in the file are separated by blank line/newline ("\n").

You **will need to submit your code and a report**. For the code, please **use the provided Jupyter notebook template**, which is also available for download on Canvas. The sections of this notebook match the sections your report should have, which are:

1. Data Management
    a. In the Jupyter Notebook:
        i. define an appropriate experimental protocol (such as k-fold, cross validation, etc);
        ii. create the dataloader to load the data.
    b. In the report:
        i. detail the experimental protocol, i.e., how you split the data;
        ii. explain your dataloader, including any pre-processing (such as encoding, vocabulary creation, etc) performed in the input data.
2. Neural Networks
    a. In the Jupyter Notebook:

i. create an efficient and robust **Recurrent Neural Network (RNN)** (such as RNN, GRU, LSTM) to tackle the problem;
ii. create an efficient and robust **Transformer Network** to solve the problem;
iii. define the necessary components to train the networks (that is, loss function, optimizers, etc);
iv. train the models;
v. for all training procedures, separately plot the loss and accuracy with respect to the epoch/iteration.

NOTE: To implement both networks, you can only use PyTorch - **no** external libraries (like transformers) can be used.

b. In the report:
   i. briefly describe your networks;
   ii. explain your decisions regarding the components (loss function, optimizers, etc);
   iii. report table with the results (using appropriate metrics). Compare the methods and define the best model.

3. Evaluate models
   a. In the Jupyter Notebook:
      i. evaluate the model (the best one you obtained in the above stage) on the appropriate set.
   b. In the report:
      i. report results for the appropriate set using **at least** 3 different metrics **suitable** for your problem. The selected metrics should capture different aspects of model performance (avoid reporting closely related measures);
      ii. plot and discuss the confusion matrix.

Submission and Team Size

For this assignment, you can work independently, in pairs, or in a group of **THREE** or **FOUR**. You are free to choose your own team members, and the size of the team will have no impact on the final mark given. Also, all members of a team will get the same grade.

The deadline is the 8th of May at 5PM.

You will need to submit your completed assessment **on Canvas**. Submit **a single zip file** which includes a report (in pdf) and the python code (in .ipynb). Your report should **not be more than 1200 words long**.

Your submission (that is, your .zip file) should have the name in the format of: "studentIDOfMember1_ studentIDOfMember2_studentIDOfMember3_studentIDOfMember4.zip" (example 201712345_201812345.zip)
Please, include the student IDs of all members. Also, all members need to make the final submission on Canvas.

Rubric

The project assessment against the following criteria:

1. Python code (**60%**)
    a. Correct definition of the data split / protocol (**5%**)
    b. Appropriate dataloader with correct encoding, vocabulary, etc (**13%**)
    c. Correct definition of network architectures as well as necessary components (**20%**)
    d. Train the models and plot graphs (**17%**)
    e. Test using the appropriate set (**5%**)
2. Report (**40%**)
    a. Clearly and briefly describe all details of the experimental protocol (**5%**)
    b. Explain the details of your dataloader, including any pre-processing performed (such as encoding, vocabulary creation, etc) (**12%**)
    c. Briefly describe the exploited network architectures, as well as the components required for training (**13%**)
    d. Report and discuss the training results table using appropriate metrics - clearly define the best model and justify/discuss (**5%**)
    e. Present and discuss a table and a confusion matrix with the final results using the best model in the appropriate set and with appropriate metrics (**5%**)